

15.6 Confidence Limits on Estimated Model Parameters

Several times already in this chapter we have made statements about the standard errors, or uncertainties, in a set of M estimated parameters $\mathbf{a}$. We have given some formulas for computing standard deviations or variances of individual parameters (equations 15.2.9, 15.4.15, and 15.4.19), as well as some formulas for covariances between pairs of parameters (equation 15.2.10; remark following equation 15.4.15; equation 15.4.20; equation 15.5.15).

In this section, we want to be more explicit regarding the precise meaning of these quantitative uncertainties, and to give further information about how quantitative confidence limits on fitted parameters can be estimated. The subject can get somewhat technical, and even somewhat confusing, so we will try to make precise statements, even when they must be offered without proof.

Figure 15.6.1 shows the conceptual scheme of an experiment that “measures” a set of parameters. There is some underlying true set of parameters $\mathbf{a}_{\text{true}}$ that are known to Mother Nature but hidden from the experimenter. These true parameters are statistically realized, along with random measurement errors, as a measured data set, which we will symbolize as $\mathcal{D}_{(0)}$. The data set $\mathcal{D}_{(0)}$ is known to the experimenter. He or she fits the data to a model by χ^2 minimization or some other technique and obtains measured, i.e., fitted, values for the parameters, which we here denote $\mathbf{a}_{(0)}$.

Because measurement errors have a random component, $\mathcal{D}_{(0)}$ is not a unique realization of the true parameters $\mathbf{a}_{\text{true}}$. Rather, there are infinitely many other realizations of the true parameters as “hypothetical data sets” each of which *could* have been the one measured, but happened not to be. Let us symbolize these by $\mathcal{D}_{(1)}, \mathcal{D}_{(2)}, \dots$. Each one, had it been realized, would have given a slightly different set of fitted parameters, $\mathbf{a}_{(1)}, \mathbf{a}_{(2)}, \dots$, respectively. These parameter sets $\mathbf{a}_{(i)}$ therefore occur with some probability distribution in the M -dimensional space of all possible parameter sets $\mathbf{a}$. The actual measured set $\mathbf{a}_{(0)}$ is one member drawn from this distribution.

Even more interesting than the probability distribution of $\mathbf{a}_{(i)}$ would be the distribution of the difference $\mathbf{a}_{(i)} - \mathbf{a}_{\text{true}}$. This distribution differs from the former one by a translation that puts Mother Nature’s true value at the origin. If we knew *this* distribution, we would know everything that there is to know about the quantitative uncertainties in our experimental measurement $\mathbf{a}_{(0)}$.

So the name of the game is to find some way of estimating or approximating the probability distribution of $\mathbf{a}_{(i)} - \mathbf{a}_{\text{true}}$ without knowing $\mathbf{a}_{\text{true}}$ and without having available to us an infinite universe of hypothetical data sets.

15.6.1 Monte Carlo Simulation of Synthetic Data Sets

Although the measured parameter set $\mathbf{a}_{(0)}$ is not the true one, let us consider a fictitious world in which it *was* the true one. Since we hope that our measured parameters are not *too* wrong, we hope that that fictitious world is not too different from the actual world with parameters $\mathbf{a}_{\text{true}}$. In particular, let us hope — no, let us *assume* — that the shape of the probability distribution $\mathbf{a}_{(i)} - \mathbf{a}_{(0)}$ in the fictitious world is the same, or very nearly the same, as the shape of the probability distribution

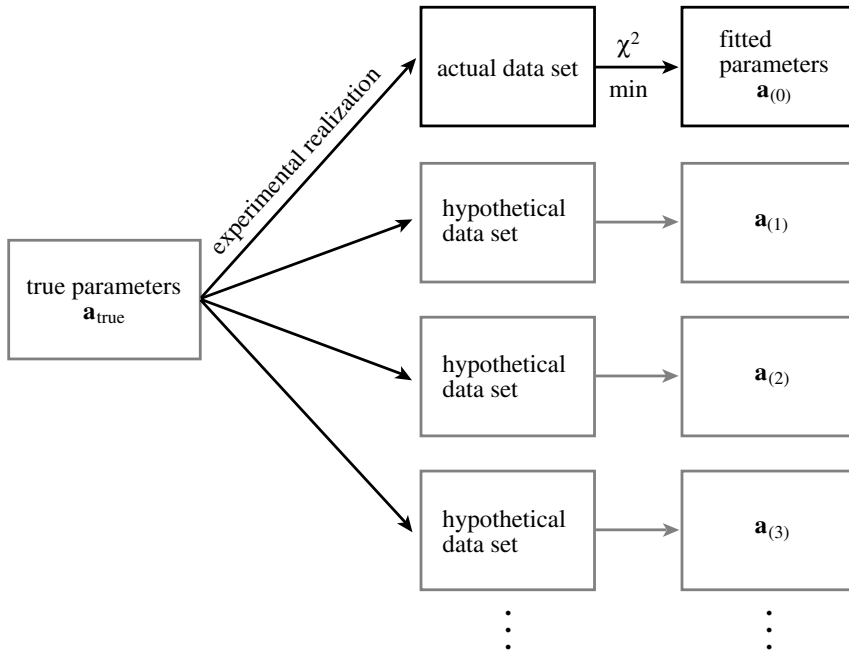


Figure 15.6.1. A statistical universe of data sets from an underlying model. True parameters $\mathbf{a}_{\text{true}}$ are realized in a data set, from which fitted (observed) parameters $\mathbf{a}_{(0)}$ are obtained. If the experiment were repeated many times, new data sets and new values of the fitted parameters would be obtained.

$\mathbf{a}_{(i)} - \mathbf{a}_{\text{true}}$ in the real world. Notice that we are not assuming that $\mathbf{a}_{(0)}$ and $\mathbf{a}_{\text{true}}$ are equal; they are certainly not. We are only assuming that the way in which random errors enter the experiment and data analysis does not vary rapidly as a function of $\mathbf{a}_{\text{true}}$, so that $\mathbf{a}_{(0)}$ can serve as a reasonable surrogate.

Now, often, the distribution of $\mathbf{a}_{(i)} - \mathbf{a}_{(0)}$ in the fictitious world *is* within our power to calculate (see Figure 15.6.2). If we know something about the process that generated our data, given an assumed set of parameters $\mathbf{a}_{(0)}$, then we can usually figure out how to *simulate* our own sets of “synthetic” realizations of these parameters as “synthetic data sets.” The procedure is to draw random numbers from appropriate distributions (cf. §7.3 – §7.4) so as to mimic our best understanding of the underlying process and measurement errors in our apparatus. With such random draws, we construct data sets with exactly the same numbers of measured points, and precisely the same values of all control (independent) variables, as our actual data set $\mathcal{D}_{(0)}$. Let us call these simulated data sets $\mathcal{D}_{(1)}^S, \mathcal{D}_{(2)}^S, \dots$. By construction, these are supposed to have exactly the same statistical relationship to $\mathbf{a}_{(0)}$ as the $\mathcal{D}_{(i)}$ ’s have to $\mathbf{a}_{\text{true}}$. (For the case where you don’t know enough about what you are measuring to do a credible job of simulating it, see below.)

Next, for each $\mathcal{D}_{(j)}^S$, perform exactly the same procedure for estimation of parameters, e.g., χ^2 minimization, as was performed on the actual data to get the parameters $\mathbf{a}_{(0)}$, giving simulated measured parameters $\mathbf{a}_{(1)}^S, \mathbf{a}_{(2)}^S, \dots$. Each simulated measured parameter set yields a point $\mathbf{a}_{(i)}^S - \mathbf{a}_{(0)}$. Simulate enough data sets and enough derived simulated measured parameters, and you map out the desired probability distribution in M dimensions.

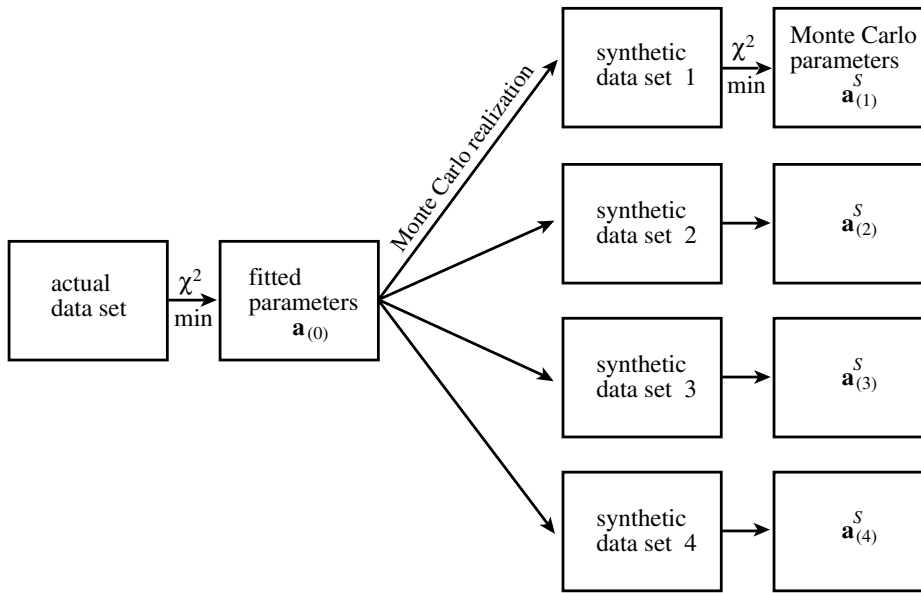


Figure 15.6.2. Monte Carlo simulation of an experiment. The fitted parameters from an actual experiment are used as surrogates for the true parameters. Computer-generated random numbers are used to simulate many synthetic data sets. Each of these is analyzed to obtain its fitted parameters. The distribution of these fitted parameters around the (known) surrogate true parameters is thus studied.

In fact, the ability to do *Monte Carlo simulations* in this fashion has revolutionized many fields of modern experimental science. Not only is one able to characterize the errors of parameter estimation in a very precise way; one can also try out on the computer different methods of parameter estimation, or different data reduction techniques, and seek to minimize the uncertainty of the result according to any desired criteria. Offered the choice between mastery of a five-foot shelf of analytical statistics books and middling ability at performing statistical Monte Carlo simulations, we would surely choose to have the latter skill.

15.6.2 Quick-and-Dirty Monte Carlo: The Bootstrap Method

Here is a powerful technique that can often be used when you don't know enough about the underlying process, or the nature of your measurement errors, to do a credible Monte Carlo simulation. Suppose that your data set consists of N *independent and identically distributed* (or *iid*) "data points." Each data point probably consists of several numbers, e.g., one or more control variables (uniformly distributed, say, in the range that you have decided to measure) and one or more associated measured values (each distributed however Mother Nature chooses). "Iid" means that the sequential order of the data points is not of consequence to the process that you are using to get the fitted parameters $\mathbf{a}$. For example, a χ^2 sum like (15.5.5) does not care in what order the points are added. Even simpler examples are the mean value of a measured quantity and the mean of some function of the measured quantities.

The *bootstrap method* [1] uses the actual data set $\mathcal{D}_{(0)}^S$, with its N data points, to generate any number of synthetic data sets $\mathcal{D}_{(1)}^S, \mathcal{D}_{(2)}^S, \dots$, also with N data points. The procedure is simply to draw N data points at a time *with replacement* from the set $\mathcal{D}_{(0)}^S$. Because of the replacement, you do not simply get back your original data set each time. You get sets in which a random fraction of the original points, typically $\sim 1/e \approx 37\%$, are replaced by *duplicated* original points. Now, exactly as in the previous discussion, you subject these data sets to the same estimation procedure as was performed on the actual data, giving a set of simulated measured parameters $\mathbf{a}_{(1)}^S, \mathbf{a}_{(2)}^S, \dots$. These will be distributed around $\mathbf{a}_{(0)}$ in close to the same way that $\mathbf{a}_{(0)}$ is distributed around $\mathbf{a}_{\text{true}}$.

Sounds like getting something for nothing, doesn't it? In fact, it took a while for the bootstrap method to become accepted by statisticians. By now, however, enough theorems have been proved to render the bootstrap reputable (see [2] for references). The basic idea behind the bootstrap is that the actual data set, viewed as a probability distribution consisting of delta functions at the measured values, is in most cases the best — or only — available estimator of the underlying probability distribution. It takes courage, but one can often simply use *that* distribution as the basis for Monte Carlo simulations.

Watch out for cases where the bootstrap's iid assumption is violated. For example, if you have made measurements at evenly spaced intervals of some control variable, then you can *usually* get away with pretending that these are iid uniformly distributed over the measured range. However, some estimators of $\mathbf{a}$ (e.g., ones involving Fourier methods) might be particularly sensitive to all the points on a grid being present. In that case, the bootstrap is going to give a wrong distribution. Also watch out for estimators that look at anything like small-scale clumpiness within the N data points, or estimators that sort the data and look at sequential differences. Obviously the bootstrap will fail on these, too. (The theorems justifying the method are still true, but some of their technical assumptions are violated by these examples.)

For a large class of problems, however, the bootstrap does yield easy, *very quick*, Monte Carlo estimates of the errors in an estimated parameter set.

15.6.3 Confidence Limits

Rather than present all details of the probability distribution of errors in parameter estimation, it is common practice to summarize the distribution in the form of *confidence limits*. The full probability distribution is a function defined on the M -dimensional space of parameters $\mathbf{a}$. A *confidence region* (or *confidence interval*) is just a region of that M -dimensional space (hopefully a small region) that contains a certain (hopefully large) percentage of the total probability distribution. You point to a confidence region and say, e.g., “there is a 99% chance that the true parameter values fall within this region around the measured value.”

It is worth emphasizing that you, the experimenter, get to pick both the *confidence level* (99% in the above example) and the shape of the confidence region. The only requirement is that your region does include the stated percentage of probability. Certain percentages are, however, customary in scientific usage: 68.3% (the lowest confidence worthy of quoting), 90%, 95.4%, 99%, and 99.73%. Higher confidence levels are conventionally “ninety-nine point nine . . . nine.” As for shape, obviously you want a region that is compact and reasonably centered on your mea-

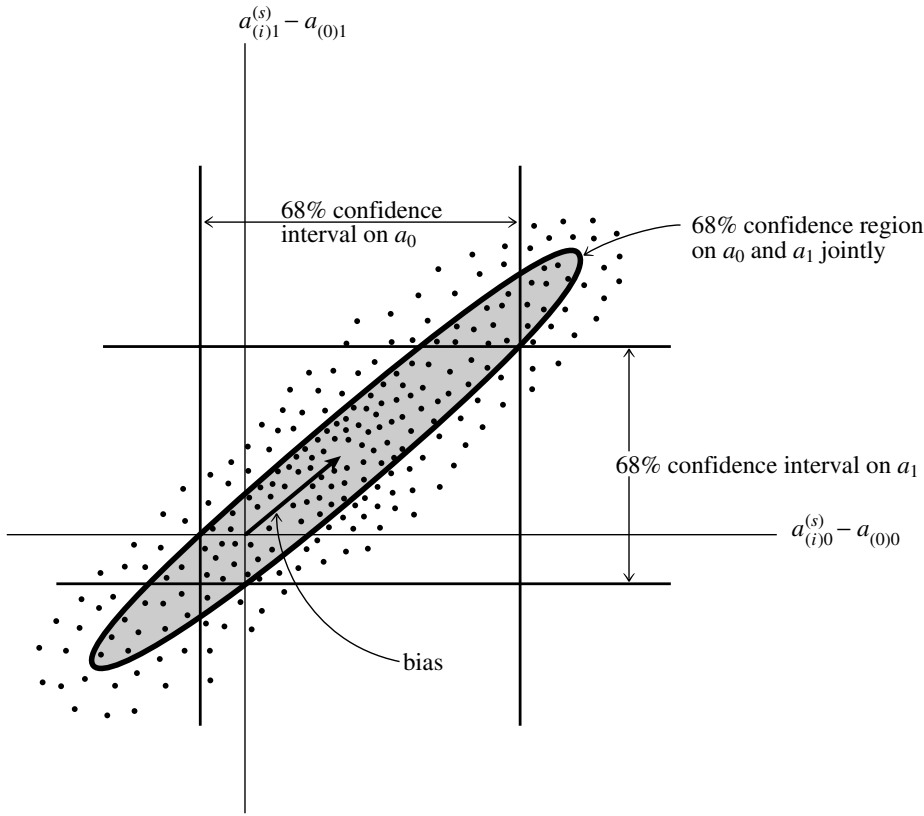


Figure 15.6.3. Confidence intervals in one and two dimensions. The same fraction of measured points (here 68%) lies (i) between the two vertical lines, (ii) between the two horizontal lines, (iii) within the ellipse.

surement $\mathbf{a}_{(0)}$, since the whole purpose of a confidence limit is to inspire confidence in that measured value. In one dimension, the convention is to use a line segment centered on the measured value; in higher dimensions, ellipses or ellipsoids are most frequently used.

You might suspect, correctly, that the numbers 68.3%, 95.4%, and 99.73%, and the use of ellipsoids, have some connection with a normal distribution. That is true historically, but not always relevant nowadays. In general, the probability distribution of the parameters will not be normal, and the above numbers, used as levels of confidence, are purely matters of convention.

Figure 15.6.3 sketches a possible probability distribution for the case $M = 2$. Shown are three different confidence regions that might usefully be given, all at the same confidence level. The two vertical lines enclose a band (horizontal interval) that represents the 68% confidence interval for the variable a_0 without regard to the value of a_1 . Similarly the horizontal lines enclose a 68% confidence interval for a_1 . The ellipse shows a 68% confidence interval for a_0 and a_1 jointly. Notice that to enclose the same probability as the two bands, the ellipse must necessarily extend outside of both of them (a point we will return to below).

15.6.4 Constant Chi-Square Boundaries as Confidence Limits

When the method used to estimate the parameters $\mathbf{a}_{(0)}$ is chi-square minimization, as in the previous sections of this chapter, then there is a natural choice for the shape of confidence intervals, whose use is almost universal. For the observed data set $\mathcal{D}_{(0)}$, the value of χ^2 is a minimum at $\mathbf{a}_{(0)}$. Call this minimum value $\chi_{\min}^2$. If the vector $\mathbf{a}$ of parameter values is perturbed away from $\mathbf{a}_{(0)}$, then χ^2 increases. The region within which χ^2 increases by no more than a set amount $\Delta\chi^2$ defines some M -dimensional confidence region around $\mathbf{a}_{(0)}$. If $\Delta\chi^2$ is set to be a large number, this will be a big region; if it is small, it will be small. Somewhere in between there will be choices of $\Delta\chi^2$ that cause the region to contain, variously, 68%, 90%, etc., of probability distribution for $\mathbf{a}$'s, as defined above. These regions are taken as the confidence regions for the parameters $\mathbf{a}_{(0)}$.

Very frequently one is interested not in the full M -dimensional confidence region, but in individual confidence regions for some smaller number ν of parameters. For example, one might be interested in the confidence interval of each parameter taken separately (the bands in Figure 15.6.3), in which case $\nu = 1$. In that case, the natural confidence regions in the ν -dimensional subspace of the M -dimensional parameter space are the *projections* of the M -dimensional regions defined by fixed $\Delta\chi^2$ into the ν -dimensional spaces of interest. In Figure 15.6.4, for the case $M = 2$, we show regions corresponding to several values of $\Delta\chi^2$. The one-dimensional confidence interval in a_1 corresponding to the region bounded by $\Delta\chi^2 = 1$ lies between the lines A and A' .

Note that it is the projection of the higher-dimensional region on the lower-dimension space that is used, not the intersection. The intersection would be the band between Z and Z' . It is *never* used. It is shown in the figure only for the purpose of making this cautionary point, that it should not be confused with the projection.

15.6.5 Probability Distribution of Parameters in the Normal Case

You may be wondering why we have, in this section up to now, made no connection at all with the error estimates that come out of the χ^2 fitting procedure, most notably the covariance matrix C_{ij} . The reason is this: χ^2 minimization is a useful means for estimating parameters even if the measurement errors are not normally distributed. While normally distributed errors are required if the χ^2 parameter estimate is to be a maximum likelihood estimator (§15.1), one is often willing to give up that property in return for the relative convenience of the χ^2 procedure. Only in extreme cases, i.e., measurement error distributions with very large “tails,” is χ^2 minimization abandoned in favor of more robust techniques, as will be discussed in §15.7.

However, the formal covariance matrix that comes out of a χ^2 minimization has a clear quantitative interpretation only if (or to the extent that) the measurement errors actually are normally distributed. In the case of *nonnormal* errors, you are “allowed”

- to fit for parameters by minimizing χ^2
- to use a contour of constant $\Delta\chi^2$ as the boundary of your confidence region
- to use Monte Carlo simulation or detailed analytic calculation in determining

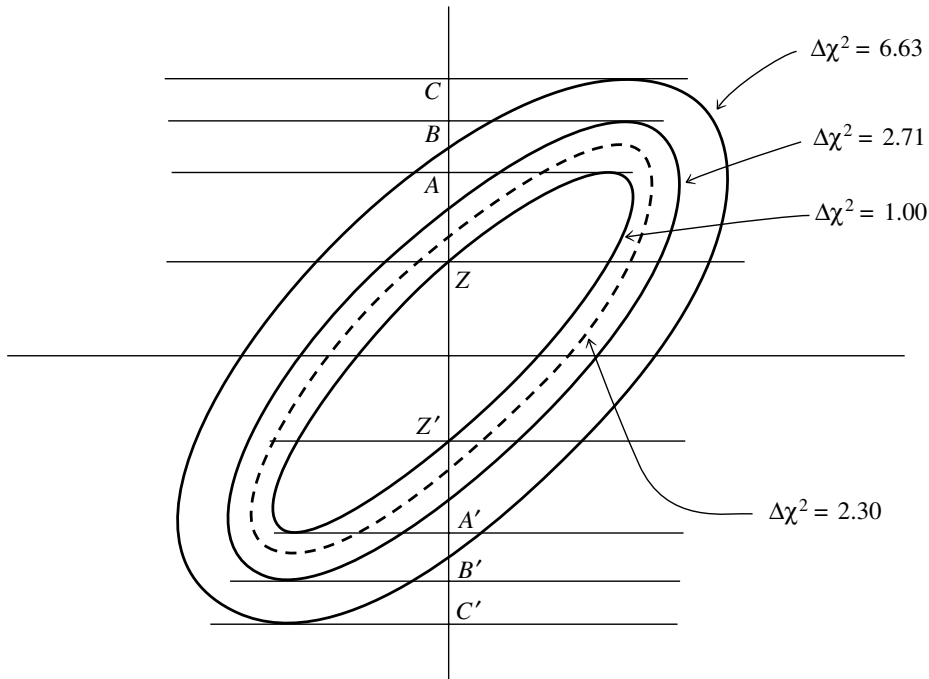


Figure 15.6.4. Confidence region ellipses corresponding to values of chi-square larger than the fitted minimum. The solid curves, with $\Delta\chi^2 = 1.00, 2.71, 6.63$, project onto one-dimensional intervals AA', BB', CC' . These intervals — not the ellipses themselves — contain 68.3%, 90%, and 99% of normally distributed data. The ellipse that contains 68.3% of normally distributed data is shown dashed and has $\Delta\chi^2 = 2.30$. For additional numerical values, see the table on p. 815.

which contour $\Delta\chi^2$ is the correct one for your desired confidence level

- to give the covariance matrix C_{ij} as the “formal covariance matrix of the fit.”

You are *not* allowed

- to use formulas that we now give for the case of normal errors, which establish quantitative relationships among $\Delta\chi^2$, C_{ij} , and the confidence level.

Here are the key theorems that hold when (i) the measurement errors are normally distributed, and either (ii) the model is linear in its parameters or (iii) the sample size is large enough that the uncertainties in the fitted parameters $\mathbf{a}$ do not extend outside a region in which the model could be replaced by a suitable linearized model. [Note that condition (iii) does not preclude your use of a nonlinear routine like `Fitmrq` to find the fitted parameters.]

Theorem A. $\chi^2_{\min}$ is distributed as a chi-square distribution with $N - M$ degrees of freedom, where N is the number of data points and M is the number of fitted parameters. This is the basic theorem that lets you evaluate the goodness-of-fit of the model, as discussed above in §15.1. We list it first to remind you that unless the goodness-of-fit is credible, the whole estimation of parameters is suspect.

Theorem B. If $\mathbf{a}_{(j)}^S$ is drawn from the universe of simulated data sets with actual parameters $\mathbf{a}_{(0)}$, then the probability distribution of $\delta\mathbf{a} \equiv \mathbf{a}_{(j)}^S - \mathbf{a}_{(0)}$ is the multivariate normal distribution

$$P(\delta \mathbf{a}) da_0 \dots da_{M-1} = \text{const.} \times \exp\left(-\frac{1}{2} \delta \mathbf{a} \cdot \boldsymbol{\alpha} \cdot \delta \mathbf{a}\right) da_0 \dots da_{M-1}$$

where $\boldsymbol{\alpha}$ is the curvature matrix defined in equation (15.5.8).

Theorem C. If $\mathbf{a}_{(j)}^S$ is drawn from the universe of simulated data sets with actual parameters $\mathbf{a}_{(0)}$, then the quantity $\Delta\chi^2 \equiv \chi^2(\mathbf{a}_{(j)}) - \chi^2(\mathbf{a}_{(0)})$ is distributed as a chi-square distribution with M degrees of freedom. Here the χ^2 's are all evaluated using the fixed (actual) data set $\mathcal{D}_{(0)}$. This theorem makes the connection between particular values of $\Delta\chi^2$ and the fraction of the probability distribution that they enclose as an M -dimensional region, i.e., the confidence level of the M -dimensional confidence region.

Theorem D. Suppose that $\mathbf{a}_{(j)}^S$ is drawn from the universe of simulated data sets (as above); that its first ν components $a_0, \dots, a_{\nu-1}$ are held fixed; and that its remaining $M - \nu$ components are varied so as to minimize χ^2 . Call this minimum value χ_v^2 . Then $\Delta\chi_v^2 \equiv \chi_v^2 - \chi_{\min}^2$ is distributed as a chi-square distribution with ν degrees of freedom. If you consult Figure 15.6.4, you will see that this theorem connects the *projected* $\Delta\chi^2$ region with a confidence level. In the figure, a point that is held fixed in a_1 and allowed to vary in a_0 minimizing χ^2 will seek out the ellipse whose top or bottom edge is tangent to the line of constant a_1 , and is therefore the line that projects it onto the smaller-dimensional space.

As a first example, let us consider the case $\nu = 1$, where we want to find the confidence interval of a single parameter, say a_0 . Notice that the chi-square distribution with $\nu = 1$ degree of freedom is the same distribution as that of the square of a single normally distributed quantity. Thus $\Delta\chi_v^2 < 1$ occurs 68.3% of the time (1- σ for the normal distribution), $\Delta\chi_v^2 < 4$ occurs 95.4% of the time (2- σ for the normal distribution), $\Delta\chi_v^2 < 9$ occurs 99.73% of the time (3- σ for the normal distribution), etc. In this manner you find the $\Delta\chi_v^2$ that corresponds to your desired confidence level. (Additional values are given in the table on the next page.)

Let $\delta \mathbf{a}$ be a change in the parameters whose first component is arbitrary, δa_0 , but the rest of whose components are chosen to minimize the $\Delta\chi^2$. Then Theorem D applies. The value of $\Delta\chi^2$ is given in general by

$$\Delta\chi^2 = \delta \mathbf{a} \cdot \boldsymbol{\alpha} \cdot \delta \mathbf{a} \quad (15.6.1)$$

which follows from equation (15.5.8) applied at $\chi_{\min}^2$ where $\beta_k = 0$. Since $\delta \mathbf{a}$ by hypothesis minimizes χ^2 in all but its zeroth component, components 1 through $M - 1$ of the normal equations (15.5.9) continue to hold. Therefore, the solution of (15.5.9) is

$$\delta \mathbf{a} = \boldsymbol{\alpha}^{-1} \cdot \begin{pmatrix} c \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \mathbf{C} \cdot \begin{pmatrix} c \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad (15.6.2)$$

where c is one arbitrary constant that we get to adjust to make (15.6.1) give the desired left-hand value. Plugging (15.6.2) into (15.6.1) and using the fact that $\mathbf{C}$ and $\boldsymbol{\alpha}$ are inverse matrices of one another, we get

$$c = \delta a_0 / C_{00} \quad \text{and} \quad \Delta\chi_v^2 = (\delta a_0)^2 / C_{00} \quad (15.6.3)$$

or

$\Delta\chi^2$ as a Function of Confidence Level p and Number of Parameters of Interest ν						
p	ν					
	1	2	3	4	5	6
68.27%	1.00	2.30	3.53	4.72	5.89	7.04
90%	2.71	4.61	6.25	7.78	9.24	10.6
95.45%	4.00	6.18	8.02	9.72	11.3	12.8
99%	6.63	9.21	11.3	13.3	15.1	16.8
99.73%	9.00	11.8	14.2	16.3	18.2	20.1
99.99%	15.1	18.4	21.1	23.5	25.7	27.9

$$\delta a_0 = \pm \sqrt{\Delta\chi^2_\nu} \sqrt{C_{00}} \quad (15.6.4)$$

At last! A relation between the confidence interval $\pm\delta a_0$ and the formal standard error $\sigma_0 \equiv \sqrt{C_{00}}$. Not unreasonably, we find that the 68% confidence interval is $\pm\sigma_0$, the 95% confidence interval is $\pm 2\sigma_0$, etc.

These considerations hold not just for the individual parameters a_i , but also for any linear combination of them: If

$$b \equiv \sum_{k=0}^{M-1} c_k a_k = \mathbf{c} \cdot \mathbf{a} \quad (15.6.5)$$

then the 68% confidence interval on b is

$$\delta b = \pm \sqrt{\mathbf{c} \cdot \mathbf{C} \cdot \mathbf{c}} \quad (15.6.6)$$

However, these simple, normal-sounding numerical relationships do *not* hold in the case $\nu > 1$ [3]. In particular, $\Delta\chi^2 = 1$ is not the boundary, nor does it project onto the boundary, of a 68.3% confidence region when $\nu > 1$. If you want to calculate not confidence intervals in one parameter, but confidence ellipses in two parameters jointly, or ellipsoids in three, or higher, then you must follow the following prescription for implementing Theorems C and D above:

- Let ν be the number of fitted parameters whose joint confidence region you wish to display, $\nu \leq M$. Call these parameters the “parameters of interest.”
- Let p be the confidence limit desired, e.g., $p = 0.68$ or $p = 0.95$.
- Find Δ (i.e., $\Delta\chi^2$) such that the probability of a chi-square variable with ν degrees of freedom being less than Δ is p . For some useful values of p and ν , Δ is given in the table above. For other values, you can use the `invcdf` method of the `Chisqdist` object in §6.14.8 with p as the argument.
- Take the $M \times M$ covariance matrix $\mathbf{C} = \boldsymbol{\alpha}^{-1}$ of the chi-square fit. Copy the intersection of the ν rows and columns corresponding to the parameters of interest into a $\nu \times \nu$ matrix denoted $\mathbf{C}_{\text{proj}}$.
- Invert the matrix $\mathbf{C}_{\text{proj}}$. (In the one-dimensional case this was just taking the reciprocal of the element C_{00} .)

- The equation for the elliptical boundary of your desired confidence region in the ν -dimensional subspace of interest is

$$\Delta = \delta \mathbf{a}' \cdot \mathbf{C}_{\text{proj}}^{-1} \cdot \delta \mathbf{a}' \quad (15.6.7)$$

where $\delta \mathbf{a}'$ is the ν -dimensional vector of parameters of interest.

If you are confused at this point, you may find it helpful to compare Figure 15.6.4 and the table on the previous page considering the case $M = 2$ with $\nu = 1$ and $\nu = 2$. You should be able to verify the following statements: (i) The horizontal band between C and C' contains 99% of the probability distribution, so it is a confidence limit on a_1 alone at this level of confidence. (ii) Ditto the band between B and B' at the 90% confidence level. (iii) The dashed ellipse, labeled by $\Delta\chi^2 = 2.30$, contains 68.3% of the probability distribution, so it is a confidence region for a_0 and a_1 jointly, at this level of confidence.

Another point of possible confusion might also be worth airing here. In §15.1.1, when we discussed the use of χ^2 as a goodness-of-fit statistic, we mentioned that a “moderately good” fit could have a χ^2 value that differed by as much as $\pm\sqrt{2\nu}$ from its expected value ν (now the total number of degrees of freedom $N - M$, not ν as used above). Indeed, the suggested tail probability that embodies this advice is $Q = 1 - \text{Chisqdist}(\nu). \text{invcdf}(\chi^2)$. Yet, in the discussion above, we seem to be saying that small changes in χ^2 , as little as ± 1 or ± 2.71 (see table on the previous page), are significant. Can both statements be true?

Yes. In §15.1.1 we were considering the variation in χ^2 over a population of hypothetical data sets with the same parameter values, $\mathbf{a}_{\text{true}}$ (cf. Figure 15.6.1). These values vary by typically $\pm\sqrt{2\nu}$. By contrast, in the discussion above, we took a single data set and held it fixed. We then asked, essentially as an exercise in propagation of errors, how much uncertainty in the fitted parameter values $\mathbf{a}_0$ was generated by the uncertainties in the data. One way to see that these are quite different concepts is to think about how they should each scale with N , the number of data points. As N gets large, χ^2 scales as N , while its variation over hypothetical data sets scales as $N^{1/2}$, essentially a random walk. Now imagine $\mathbf{a}$ varying around its fitted value $\mathbf{a}_0$ by a small amount, $\mathbf{a} = \mathbf{a}_0 + \delta \mathbf{a}$. The change in χ^2 scales with the number of terms in the sum, N , and quadratically with distance from the minimum,

$$\delta\chi^2 \propto N(\delta \mathbf{a})^2 \quad (15.6.8)$$

As the number of data points increases, we reasonably expect the parameters to become more accurately determined, scaling as

$$\delta \mathbf{a} \propto N^{-1/2} \quad (15.6.9)$$

Combining these two equations, we find that $\delta\chi^2$ for the minimum significant change in parameters $\delta \mathbf{a}$ scales as N^0 , that is, as a constant. In fact, Theorems B and C above tell us that this is not just reasonable expectation on our part; it is actually true.

15.6.6 Confidence Limits from Singular Value Decomposition

When you have obtained your χ^2 fit by singular value decomposition (§15.4), the information about the fit’s formal errors comes packaged in a somewhat different,

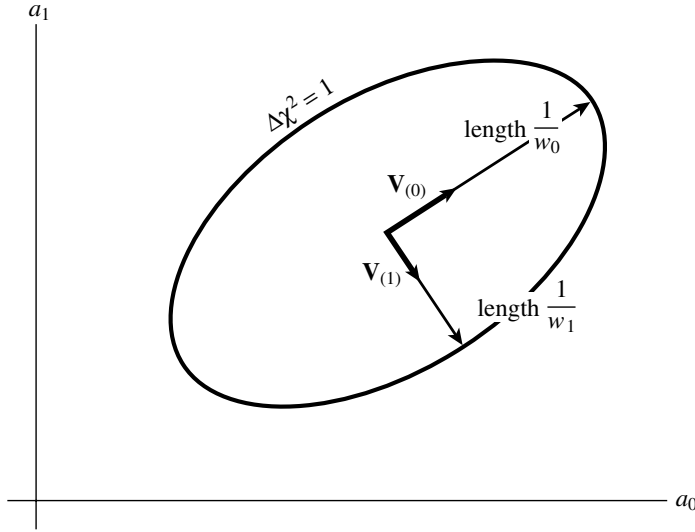


Figure 15.6.5. Relation of the confidence region ellipse $\Delta\chi^2 = 1$ to quantities computed by singular value decomposition. The vectors $\mathbf{V}_{(i)}$ are unit vectors along the principal axes of the confidence region. The semi-axes have lengths equal to the reciprocal of the singular values w_i . If the axes are all scaled by some constant factor α , $\Delta\chi^2$ is scaled by the factor α^2 .

but generally more convenient, form. The columns of the matrix $\mathbf{V}$ are an orthonormal set of M vectors that are the principal axes of the $\Delta\chi^2 = \text{constant}$ ellipsoids. We denote the columns as $\mathbf{V}_{(0)} \dots \mathbf{V}_{(M-1)}$. The lengths of those axes are inversely proportional to the corresponding singular values $w_0 \dots w_{M-1}$; see Figure 15.6.5. The boundaries of the ellipsoids are thus given by

$$\Delta\chi^2 = w_0^2(\mathbf{V}_{(0)} \cdot \delta\mathbf{a})^2 + \dots + w_{M-1}^2(\mathbf{V}_{(M-1)} \cdot \delta\mathbf{a})^2 \quad (15.6.10)$$

which is the justification for writing equation (15.4.18) above. Keep in mind that it is *much* easier to plot an ellipsoid given a list of its vector principal axes than given its matrix quadratic form: Loop over points $\mathbf{z}$ on a unit sphere in any desired way (e.g., by latitude and longitude) and plot the mapped points

$$\delta\mathbf{a} = \sqrt{\Delta\chi^2} \sum_i \frac{1}{w_i} (\mathbf{z} \cdot \mathbf{V}_{(i)}) \mathbf{V}_{(i)} \quad (15.6.11)$$

The formula for the covariance matrix $\mathbf{C}$ in terms of the columns $\mathbf{V}_{(i)}$ is

$$\mathbf{C} = \sum_{i=0}^{M-1} \frac{1}{w_i^2} \mathbf{V}_{(i)} \otimes \mathbf{V}_{(i)} \quad (15.6.12)$$

or, in components,

$$C_{jk} = \sum_{i=0}^{M-1} \frac{1}{w_i^2} V_{ji} V_{ki} \quad (15.6.13)$$

A method for plotting error ellipses (2-dimensions) or ellipsoids (3-dimensions) from the covariance matrix $\mathbf{C}$ directly, not using its principal axes, is described in §16.1.1.